

PROJECT REPORT

AI & ML Internship Task 6

Project Title: Data Cleaning and Preprocessing on Telecom Churn Dataset

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5. Introduction

5.1 What is Data Cleaning

Data cleaning is the initial and arguably most vital phase of any data science pipeline. It involves identifying and correcting inaccuracies, inconsistencies, and errors within a raw dataset. Real-world data is often noisy, incomplete, or incorrectly formatted due to human error during data entry, system glitches, or integration issues from multiple sources. Data cleaning addresses missing values, duplicate entries, structural errors, and unreadable formats, ensuring that the underlying data is accurate and reliable. The fundamental goal is to create a reliable foundation upon which analytical models can be built without the risk of propagating systemic errors.

5.2 What is Data Preprocessing

While data cleaning focuses on correctness, data preprocessing focuses on transformation. It is the process of converting raw data into an understandable, well-structured format that machine learning algorithms can ingest. This involves techniques such as scaling numerical features so they share a common magnitude, encoding categorical variables into mathematical representations, handling outliers that could skew model weights, and engineering new features that better capture the underlying business logic. Preprocessing acts as the critical bridge between raw data extraction and model training.

5.3 Importance in Machine Learning

The foundational principle of machine learning is "Garbage In, Garbage Out" (GIGO). An algorithm's ability to learn complex patterns and make accurate predictions is strictly bound by the quality of the data it is trained on. Unscaled data can cause algorithms like K-Nearest Neighbors or Support Vector Machines to perform poorly. Unhandled categorical data will crash mathematical models, and outliers can drastically shift regression lines. Proper preprocessing reduces model training time, improves overall accuracy, and prevents overfitting, making it an indispensable skill for production-ready AI models.

5.4 Telecom Churn Prediction

The telecommunications industry operates in a highly competitive market where the cost of acquiring a new customer significantly outweighs the cost of retaining an existing one. "Churn" refers to the phenomenon where a customer cancels their subscription and moves to a competitor. Predicting churn allows telecom companies to proactively offer incentives to at-risk customers. For a predictive model to accurately identify these customers based on their usage patterns and demographics, the historical data must first be rigorously cleaned and standardized.

6. Problem Statement

Raw datasets generated by telecom operating systems contain a mixture of numerical, categorical, and datetime formats. They often suffer from disparate scales, extreme outliers in usage metrics, and non-machine-readable text categories. The objective of this project is to intake the raw `telecom_churn.csv` dataset, diagnose its structural and statistical anomalies, and systematically apply data cleaning and preprocessing techniques to output a refined, machine-learning-ready file (`preprocessed_dataset.csv`). The challenge lies in ensuring that the data transformations do not introduce bias or lead to information loss, while simultaneously satisfying the strict numerical requirements of downstream machine learning classification algorithms.

7. Project Objectives

The primary objectives of Task 6 are as follows:

- **Data Ingestion & Inspection:** To successfully load the dataset and analyze its structural dimensions, data types, and initial statistical summaries.
- **Anomaly Resolution:** To systematically verify the presence of missing values (Null/NaN) and duplicate records, taking corrective action if required.
- **Data Transformation:** To convert categorical text features into numerical formats using Label Encoding without losing data cardinality.
- **Outlier Management:** To visually detect extreme statistical deviations using boxplots and mathematically cap them utilizing the Interquartile Range (IQR) method.
- **Feature Engineering:** To extract meaningful, model-friendly numerical components from datetime strings to improve model interpretability.
- **Standardization:** To scale all continuous variables to a standard normal distribution, ensuring uniform influence across all features during model training.

8. Dataset Description

8.1 Dataset Overview

The dataset utilized for this task is the Telecom Churn Dataset, sourced publicly from Kaggle. It contains historical records of telecommunication customers, detailing their demographic information, geographical location, network usage metrics, and whether they ultimately churned. The dataset consists of 243,553 total records and 14 independent/dependent features.

8.2 Features Description Table

Feature Name	Data Type	Description
customer_id	Integer	Unique identifier for each telecom customer.
telecom_partner	Categorical	The specific telecom service provider.
gender	Categorical	Gender of the customer (M/F).
age	Integer	Age of the customer in years.
state	Categorical	The Indian state where the customer resides.
city	Categorical	The city where the customer resides.
pincode	Integer	The postal code of the customer's location.
date_of_registration	Datetime	The date the customer registered for the telecom service.
num_dependents	Integer	Number of dependents the customer has.
estimated_salary	Float	The estimated annual salary of the customer.
calls_made	Integer	Total number of calls made by the customer.
sms_sent	Integer	Total number of SMS messages sent.
data_used	Integer/Float	Amount of internet data utilized by the customer.
churn	Integer (Binary)	Target variable indicating if the customer churned (1) or not (0).

8.3 Target Variable Description

The target variable for future modeling is churn. It is a binary classification indicator where 1 represents a customer who has left the telecom provider, and 0 represents a customer who has been retained. The objective of downstream modeling will be to predict this variable based on the other 13 features.

9. Tools and Technologies Used

To execute the data preprocessing pipeline efficiently, the following tech stack was utilized:

- Python: The core programming language used for scripting the data manipulation logic.
- Pandas: A powerful library used for data manipulation, ingestion, structural reshaping, and dataframe operations.
- NumPy: Utilized for high-performance mathematical computations and conditional logic arrays.
- Matplotlib & Seaborn: Data visualization libraries employed to plot statistical graphics, primarily used here for generating boxplots to detect outliers.
- Scikit-Learn (sklearn): The premier machine learning library in Python. Its preprocessing module was used extensively for LabelEncoder and StandardScaler.
- Google Colab / Jupyter Notebook: The interactive computing environment utilized to write, debug, and execute the Python code sequentially.

10. Methodology

The following structured steps were executed to transform the raw dataset into a preprocessed state.

10.1 Dataset Loading

The initial step involved loading the raw CSV file into a Pandas DataFrame for manipulation in memory. This provides a tabular data structure that is highly optimized for querying and transformation.

```
df = pd.read_csv("telecom_churn.csv")
```

10.2 Data Inspection

To understand the schema, data types, and initial statistical distributions, built-in Pandas methods were utilized to extract metadata and summary statistics.

```
df.info()  
df.describe()
```

10.3 Missing Value Handling

Machine learning models cannot process incomplete data matrices. The dataset was scanned for missing (NaN) values across all columns.

```
df.isnull().sum()
```

Result: No missing values were found in any column, meaning imputation (such as KNN or mean imputation) was not required.

10.4 Duplicate Handling

Duplicate rows can bias the model by artificially inflating the importance of certain data points. The dataset was examined for exact row matches.

```
df.duplicated().sum()
```

10.5 Categorical Encoding

Algorithms require numerical input. The categorical string columns were converted into distinct integer representations using Scikit-Learn's LabelEncoder.

```
cat_cols = df.select_dtypes(include='object').columns
le = LabelEncoder()
for col in cat_cols:
    df[col] = le.fit_transform(df[col].astype(str))
```

10.6 Outlier Detection

Visual inspection of data distribution is critical. Seaborn was used to plot boxplots to identify data points that fell significantly outside the normal distribution whiskers.

```
sns.boxplot(data=df)
plt.show()
```

10.7 Outlier Treatment

To prevent extreme values from distorting future model training, the Interquartile Range (IQR) method was applied. Values outside the $1.5 * \text{IQR}$ boundaries were capped rather than deleted to preserve data volume.

```
for col in num_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    df[col] = np.where(df[col] > upper, upper, np.where(df[col] <
lower, lower, df[col]))
```

10.8 Feature Engineering

The date_of_registration column is a complex string/datetime object. It was decomposed into continuous integer variables representing the year and month, which models can interpret as seasonal or chronological trends.

```
df['date_of_registration'] =
pd.to_datetime(df['date_of_registration'])
df['registration_year'] = df['date_of_registration'].dt.year
df['registration_month'] = df['date_of_registration'].dt.month
df.drop('date_of_registration', axis=1, inplace=True)
```

10.9 Feature Scaling

To ensure that high-magnitude features do not overpower low-magnitude features, Standardization (Z-score normalization) was applied.

```
scaler = StandardScaler()  
df[num_cols] = scaler.fit_transform(df[num_cols])
```

11. Implementation and Results

The programmatic execution of the methodology yielded successful data transformation. Below are observational confirmations of the pipeline's effectiveness.

Missing Value Analysis

Upon executing `df.isnull().sum()`, the console output confirmed that zero missing values were present across all 14 features. This indicated high initial data fidelity regarding completeness, allowing the pipeline to bypass complex imputation algorithms and move directly to encoding and structural transformations.

Outlier Detection and Treatment

The visual representation generated by Seaborn highlighted significant univariate outliers, particularly in numerical attributes such as `pincode`, `estimated_salary`, and `data_used`. These outliers were subsequently treated using the mathematical IQR boundary method. By capping these values, the variance of the dataset was stabilized without losing valuable rows of customer data.

Final Preprocessed Dataset

The preview of the final exported dataframe shows a completely machine-learning-ready dataset. All text strings have been mapped to integers, the datetime object has been split into `registration_year` and `registration_month`, and the numerical values have been transformed into standard normal scales. The final processed dataset was saved as `preprocessed_dataset.csv`.

12. Challenges Faced

During the execution of this internship task, several practical challenges were encountered and successfully resolved:

- **Understanding the Preprocessing Workflow:** Initially, determining the correct sequential order of operations (e.g., encoding before scaling) required research and logical planning to avoid data leakage or incorrect transformations.
- **Working with Large Datasets:** Processing a dataframe with over 243,000 records required memory-efficient code. Iterating through rows was avoided in favor of vectorized Pandas and NumPy operations, which drastically reduced computation time.
- **Identifying Outliers:** Differentiating between genuine anomalies and natural extreme variations in a telecom setting was challenging. The IQR method was chosen over outright deletion to preserve the statistical representation.
- **Feature Transformation:** Handling the datetime object required specific Pandas syntax to extract the year and month correctly without corrupting the index.
- **Data Scaling:** Applying the StandardScaler to the dataset required ensuring that categorical encoded targets (like the churn column) were handled carefully so as not to lose their distinct binary meaning.

13. Learning Outcomes

Completing this task yielded substantial practical knowledge and technical proficiency in the following areas:

- **Data Cleaning:** Gained hands-on experience inspecting large dataframes, verifying structural integrity, and utilizing Pandas functions to assert data quality.
- **Label Encoding:** Learned how to transform human-readable string labels into machine-readable integer arrays using Scikit-Learn.
- **Outlier Treatment:** Deepened mathematical understanding of quartiles and interquartile ranges. Learned how to implement logical capping matrices utilizing NumPy vectors.
- **Feature Engineering:** Successfully manipulated complex datatypes (datetime) to create new, simplified numerical features that can improve predictive model accuracy.
- **Standardization:** Understood the mathematical mechanics behind the StandardScaler and its critical role in optimizing gradient descent in future machine learning models.
- **End-to-End Pipeline Creation:** Most importantly, learned how to architect a complete, cohesive data preparation pipeline that takes raw, messy data and systematically outputs a robust, production-ready dataset.

14. Conclusion

Data preprocessing is the backbone of successful AI and ML model development. Through the structured execution of Task 6, a raw, multi-format telecom dataset containing nearly a quarter of a million records was successfully transformed. By verifying data completeness, mapping categorical strings to numeric values, intelligently capping extreme statistical outliers, and scaling distinct variables into a uniform metric space, the dataset's overall quality and predictive viability were drastically enhanced.

The resulting `preprocessed_dataset.csv` is now highly optimized. The standard normal distribution of its features ensures that future algorithms—whether they be Logistic Regression, Random Forests, or Neural Networks—will train faster, converge more reliably, and yield a significantly higher accuracy when attempting to predict customer churn. This project successfully demonstrated that a rigorous, methodical approach to data engineering is just as vital as the machine learning models themselves.

15. References

Kaggle dataset (Telecom Churn Dataset):

<https://www.kaggle.com/datasets/suraj520/telecom-churn-dataset>

Colab Code file:

https://colab.research.google.com/drive/1Dzrh7EupG_FMEAyvNVKhzGMj7YNIBifb?usp=sharing

Preprocessed dataset:

<https://github.com/dbmodi2311/Data-Collection-Data-Source-Identification/tree/main/Task%206>